



# Storm Sequence Generator — Test Results

Automated Test Documentation for Model Governance

MKM Research Labs

Version 1.0 — 23-March-2026

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# 1 Test Results — Storm Sequence Generator (MKM-SS-001)

Tests run on **23 March 2026 at 12:58:50**.

Table 1: Test Results Summary — Storm Sequence Generator (MKM-SS-001)

|             |       |
|-------------|-------|
| Total Tests | 52    |
| Passed      | 52    |
| Failed      | 0     |
| Skipped     | 0     |
| Duration    | 0.21s |

Table 2: Detailed Test Results — Storm Sequence Generator

| Test Description                                               | Acceptance Criteria                                                                                                                | Result | Time   |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Duration must not exceed MAX_PRECIP_END_HOUR (156).            | <code>result.storms[0].duration_hours &lt;= MAX_PRECIP_END_HOUR</code>                                                             | Pass   | 0.000s |
| Isolated has one storm                                         | <code>isinstance(result, StormSequence);</code><br><code>result.num_storms == 1 (+2 more)</code>                                   | Pass   | 0.000s |
| All storm sequences                                            | <code>isinstance(seq, StormSequence)</code>                                                                                        | Pass   | 0.000s |
| Catchment propagated                                           | <code>seq.catchment_id == "rhine";</code><br><code>storm.catchment_id == "rhine"</code>                                            | Pass   | 0.001s |
| Correct count                                                  | <code>len(small_batch) == 200</code>                                                                                               | Pass   | 0.000s |
| Different seeds differ                                         | <code>types_a != types_b</code>                                                                                                    | Pass   | 0.002s |
| Force sequence type doublet                                    | <code>doublets &gt; 50; seq.sequence_type in ("doublet", "isolated")</code>                                                        | Pass   | 0.002s |
| Force sequence type isolated                                   | <code>seq.sequence_type == "isolated";</code><br><code>seq.num_storms == 1</code>                                                  | Pass   | 0.001s |
| Reproducible with same seed                                    | <code>[s.sequence_id for s in a] != [s.sequence_id for s in b];</code><br><code>sa.num_storms == sb.num_storms</code><br>(+1 more) | Pass   | 0.003s |
| Returns list                                                   | <code>isinstance(small_batch, list)</code>                                                                                         | Pass   | 0.000s |
| Custom weight heavy on extreme shifts the distribution.        | <code>all(c == "extreme" for c in cats)</code>                                                                                     | Pass   | 0.007s |
| All four default weight categories appear in the batch.        | <code>cat in categories</code>                                                                                                     | Pass   | 0.000s |
| Multi-storm sequences appear at a meaningful rate for severe+. | <code>multi_frac &gt; 0.15</code>                                                                                                  | Pass   | 0.000s |
| Creates parent dirs                                            | <code>out.exists()</code>                                                                                                          | Pass   | 0.007s |
| Load missing file raises                                       | —                                                                                                                                  | Pass   | 0.000s |
| Load returns correct count                                     | <code>len(loaded) == len(small_batch)</code>                                                                                       | Pass   | 0.020s |

| Test Description                                | Acceptance Criteria                                                                                                                | Result | Time   |
|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Load wrong schema raises                        | —                                                                                                                                  | Pass   | 0.002s |
| Loaded sequences still valid                    | <code>result["invalid"] == 0</code>                                                                                                | Pass   | 0.019s |
| Round trip num storms                           | <code>orig.num_storms == back.num_storms</code>                                                                                    | Pass   | 0.020s |
| Round trip precipitation                        | <code>abs(orig.total_precipitation_mm -<br/>back.total_precipitatio...</code>                                                      | Pass   | 0.020s |
| Round trip sequence type                        | <code>orig.sequence_type ==<br/>back.sequence_type</code>                                                                          | Pass   | 0.019s |
| Round trip storm fields                         | <code>os_.storm_index == bs_.storm_index;<br/>abs(os_.duration_hours -<br/>bs_.duration_hours) &lt; 0.01</code> (+1 more)          | Pass   | 0.021s |
| Round trip total duration                       | <code>abs(orig.total_duration_hours -<br/>back.total_duration_hours...</code>                                                      | Pass   | 0.020s |
| Save creates file                               | <code>out.exists()</code>                                                                                                          | Pass   | 0.007s |
| Schema version in file                          | <code>raw["schema_version"] ==<br/>SCHEMA_VERSION</code>                                                                           | Pass   | 0.021s |
| Creates file                                    | <code>out.exists()</code>                                                                                                          | Pass   | 0.000s |
| Duration stats present                          | <code>d["min"] &gt; 0; d["max"] &lt;= 156</code> (+1 more)                                                                         | Pass   | 0.001s |
| Num sequences matches                           | <code>data["num_sequences"] ==<br/>len(small.batch)</code>                                                                         | Pass   | 0.001s |
| Precipitation stats present                     | <code>p["min"] &gt; 0; p["max"] &gt;= p["min"]</code> (+1 more)                                                                    | Pass   | 0.002s |
| Schema version                                  | <code>data["schema_version"] ==<br/>SCHEMA_VERSION</code>                                                                          | Pass   | 0.002s |
| Type counts sum to total                        | <code>sum(data["sequence_type_counts"].values())<br/>== len(small.b...</code>                                                      | Pass   | 0.001s |
| Type fractions sum to one                       | <code>abs(total_frac - 1.0) &lt; 0.001</code>                                                                                      | Pass   | 0.001s |
| Aggregate stats                                 | <code>result.max_intensity_factor &gt;=<br/>result.avg_intensity_factor;<br/>result.total_precipitation_mm &gt; 0</code> (+1 more) | Pass   | 0.000s |
| Catchment id in storms                          | <code>result.storms[0].catchment_id ==<br/>"rhine"</code>                                                                          | Pass   | 0.000s |
| Cluster sequence                                | <code>result.sequence_type in<br/>(SequenceType.CLUSTER.value, Sequ...</code>                                                      | Pass   | 0.000s |
| force_sequence_type='doublet' → always doublet. | <code>result.sequence_type ==<br/>SequenceType.DOUBLET.value;<br/>result.num_storms == 2</code>                                    | Pass   | 0.000s |

| Test Description                                                      | Acceptance Criteria                                                                                                                  | Result | Time   |
|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| force_sequence_type='isolated' → always isolated.                     | <code>result.sequence_type ==</code><br><code>SequenceType.ISOLATED.value;</code><br><code>result.num_storms == 1</code>             | Pass   | 0.000s |
| Minimal intensity (5% sequence probability) mostly produces isolated. | <code>isolated &gt;= 10</code>                                                                                                       | Pass   | 0.000s |
| Generate multiple sequences without error.                            | <code>isinstance(result, StormSequence);</code><br><code>result.num_storms &gt;= 1</code>                                            | Pass   | 0.000s |
| Persistent sequence                                                   | <code>result.sequence_type in</code><br><code>(SequenceType.PERSISTENT.value,</code><br><code>S...</code>                            | Pass   | 0.000s |
| Returns storm sequence                                                | <code>isinstance(result, StormSequence)</code>                                                                                       | Pass   | 0.000s |
| All sequences must have total_duration ≤ 168h.                        | <code>result.total_duration_hours &lt;= 168.0</code>                                                                                 | Pass   | 0.000s |
| Storms have correct fields                                            | <code>storm.duration_hours &gt; 0;</code><br><code>storm.intensity_factor &gt; 0</code> (+2 more)                                    | Pass   | 0.000s |
| Custom catchment                                                      | <code>gen.catchment_id == "rhine";</code><br><code>gen.base_precipitation ==</code><br><code>CATCHMENT_BASE_PRECIP["rhine"]</code>   | Pass   | 0.000s |
| Default catchment                                                     | <code>gen.catchment_id == "thames";</code><br><code>gen.base_precipitation ==</code><br><code>CATCHMENT_BASE_PRECIP["thames"]</code> | Pass   | 0.000s |
| Same seed → same sequence.                                            | <code>seq1.num_storms == seq2.num_storms;</code><br><code>seq1.sequence_type ==</code><br><code>seq2.sequence_type</code>            | Pass   | 0.000s |
| Unknown catchment defaults to 35                                      | <code>gen.base_precipitation == 35.0</code>                                                                                          | Pass   | 0.000s |
| Zero validation errors across the whole batch.                        | <code>result["invalid"] == 0</code>                                                                                                  | Pass   | 0.000s |
| Medium batch all valid                                                | <code>result["invalid"] == 0</code>                                                                                                  | Pass   | 0.001s |
| No negative durations                                                 | <code>storm.duration_hours &gt; 0</code>                                                                                             | Pass   | 0.000s |
| No negative precipitation                                             | <code>seq.total_precipitation_mm &gt; 0;</code><br><code>storm.precipitation_mm &gt; 0</code>                                        | Pass   | 0.000s |
| Last storm precipitation must end by hour 156 (1e-9 FP tolerance).    | <code>last_end &lt;= MAX_PRECIP_END_HOUR +</code><br><code>_FP_TOL</code>                                                            | Pass   | 0.000s |

## 1.1 Test Document History

| Date        | Version | Description                    | Author   |
|-------------|---------|--------------------------------|----------|
| 12-Mar-2026 | 1.0     | Initial test suite (116 tests) | D. Kelly |
| 23-Mar-2026 | 1.1     | 64 test(s) removed (total: 52) | D. Kelly |